

AI-SA-IN-TH-000005

Module 8.1 · LLM Evaluation Fundamentals · AI Safety: Evaluation, Guardrails & Responsible Deployment

Human Evaluation Methods

Pairwise preference, Likert scales, inter-annotator agreement, and Chatbot Arena

8 min

Duration

B3 – Apply

Bloom's

L2 – Intermediate

Level

Theory

Domain

Learning Objectives

- 1 Explain why human evaluation remains necessary despite advances in automated and LLM-as-judge methods
- 2 Design a pairwise preference study with appropriate instructions and quality controls
- 3 Apply Likert scales correctly for multi-dimensional LLM quality assessment
- 4 Compute Cohen's kappa and Fleiss' kappa and interpret inter-annotator agreement scores
- 5 Describe the Chatbot Arena ELO leaderboard methodology and its advantages over static benchmarks
- 6 Select the appropriate human evaluation method given task type, budget, and required validity

Why Human Evaluation Remains Irreplaceable

Human Evaluation

The process of having trained human annotators, domain experts, or representative end-users assess the quality of LLM outputs according to defined criteria. Despite its cost and scaling limitations, human evaluation is the only method that directly measures what actually matters: whether the output is useful and appropriate for real humans.

- Automated metrics proxy human judgment — they are never its substitute; the chain of validation must end with humans
- Required for safety-critical domains: medical, legal, financial, educational — no automated metric can certify safety
- Essential for detecting subtle alignment failures: sycophancy, evasion, cultural insensitivity, or misleading framing
- Provides the calibration signal that makes LLM-as-judge systems reliable — without human labels, judges cannot be validated

LLM-as-Judge vs Human Evaluation — When to Use Each

LLM-as-Judge

- High volume (>10,000 evaluations): cost effective
- Rapid iteration during development — hourly if needed
- Tasks with clear, learnable rubrics: coding, factual Q&A, summarisation
- RLHF preference data generation at scale
- Continuous monitoring in production: anomaly detection
- Cannot certify safety, alignment, or cultural appropriateness

Human Evaluation

- Final deployment decision: any model touching real users
- Safety-critical domains: medicine, law, finance, education
- Subtle quality dimensions: nuance, tone, empathy, humour
- Calibrating and validating LLM judges before deployment
- Novel tasks with no established rubric or reference outputs
- Ground truth for benchmarks: all gold-standard labels are human

↳ for design system prompts

Pairwise Preference Studies

Pairwise Preference Annotation

Annotators are shown two model responses to the same prompt and asked to indicate which response they prefer (or if they are equivalent). Aggregated across many annotators and prompts, pairwise preferences produce a win-rate matrix that directly measures relative model quality on the dimensions that matter to users.

- Direct measurement of comparative quality: avoids the scale-anchoring issues of absolute rating
- Connects directly to real-world model selection: which model would you actually use?
- Inter-annotator agreement is typically higher for pairwise (clearer judgment) than for absolute Likert scales
- Annotation interface design matters: present both responses simultaneously; randomise order; limit to one dimension per study
- Sample size: 200–500 unique prompts $\times$ 2–3 annotators per prompt provides stable win-rate estimates ($\pm 3\%$ margin)



Pairwise Annotation — Mock Task Instructions

Annotation interface shown to crowd annotators (e.g. Scale AI, Surge, or Prolific platform)

Annotator Instructions

"You will be shown a question and two AI assistant responses (Response A and Response B). Read both responses carefully, then answer: Which response better answers the question?"

Select: Response A is better / Response B is better / About the same

Then select ALL reasons for your choice from the checklist:

- ☐ More accurate / factually correct
- ☐ More helpful for the task
- ☐ Clearer and easier to understand
- ☐ More complete
- ☐ Better tone / more appropriate
- ☐ Safer / more appropriate"

Aim for ~3 minutes per annotation. If unsure, choose 'About the same'.

Quality Controls

Include 5% gold-standard validation prompts with known correct responses to detect inattentive annotators. Screen out annotators with <80% agreement on gold prompts. Minimum 2 annotations per prompt; use majority vote for final label.

Likert Scale Annotation

Likert Scale Evaluation

Annotators rate a single model response on an ordinal scale (typically 1–5 or 1–7) for each quality dimension independently. The scale provides a structured absolute quality signal across multiple criteria simultaneously — useful for multi-dimensional profiling of a single model.

- Typical dimensions for LLM evaluation: Accuracy (1–5), Helpfulness (1–5), Coherence (1–5), Safety (1–5), Completeness (1–5)
- 5-point scales are standard: 1=Strongly disagree, 3=Neutral, 5=Strongly agree
- Use bipolar anchors with clear behavioural descriptions at each point — avoid 'good/bad' as anchors
- Aggregate across annotators using mean score per dimension; report standard deviation to show annotator spread
- Limitation: susceptible to scale bias — annotators from different cultures have different baselines for 'neutral'

Task-Completion Evaluation — High-Stakes Domain Study

Case: Medical Information Chatbot — Patient-Facing Deployment Evaluation

- 50 simulated patient queries across 10 medical domains (oncology, cardiology, diabetes, mental health, etc.)
- 3 annotators per query: 1 physician, 1 patient advocate, 1 layperson — representing different stakes
- Each response evaluated on: Medical Accuracy (1–5), Appropriate Referral Advice (1–5), Safety (1–5), Clarity for Patient (1–5)
- Pass/fail gating: any response with Safety < 4 OR Medical Accuracy < 3 triggers automatic failure — blocks deployment
- Inter-annotator agreement measured with Fleiss' κ ; physician $\kappa = 0.71$, patient advocate $\kappa = 0.63$, layperson $\kappa = 0.58$

Key Lesson

No LLM judge can substitute for physician review on medical content. The human evaluation revealed 3/50 (6%) responses with plausible but medically incorrect information that automated metrics did not detect. The deployment was delayed 3 weeks for model retraining.

Cohen's Kappa — Inter-Annotator Agreement

$$\kappa = (P_o - P_e) / (1 - P_e)$$

1. P_o = observed agreement = fraction of items where the two annotators gave the same label
2. P_e = expected chance agreement = $\sum (p_a \times p_b)$ for each category a and b
3. Example: 100 responses rated as (Good/Bad). Annotator 1: 60 Good, 40 Bad. Annotator 2: 65 Good, 35 Bad
4. Both Good: 42. Both Bad: 30. $P_o = (42+30)/100 = 0.72$ *≈ 1 Perfect*
5. $P_e = (60/100 \times 65/100) + (40/100 \times 35/100) = 0.39 + 0.14 = 0.53$
6. $\kappa = (0.72 - 0.53) / (1 - 0.53) = 0.19 / 0.47 = 0.40$ (Moderate agreement)

60 g 40.

Note: $\kappa < 0.2$: slight; 0.2–0.4: fair; 0.4–0.6: moderate; 0.6–0.8: substantial; > 0.8 : almost perfect

A1 60) 40
A2 65 / 35

Kappa Interpretation — Worked Annotation Scenario

Three annotators evaluate 200 LLM responses for Helpfulness (1–5 scale, collapsed to Low/Medium/High)

Raw Agreement Matrix (Annotator 1 vs Annotator 2)

A1\A2	Low	Medium	High	
Low	28	8	2	→ 38 Low
Medium	6	75	9	→ 90 Medium
High	2	8	62	→ 72 High

$$P_o = (28+75+62)/200 = 0.825$$

Kappa Calculation

$$P_e(\text{Low}) = (38/200) \times (36/200) = 0.0342$$

$$P_e(\text{Medium}) = (90/200) \times (88/200) = 0.1980$$

$$P_e(\text{High}) = (72/200) \times (76/200) = 0.1368$$

$$P_e \text{ total} = 0.369$$

$$\kappa = (0.825 - 0.369) / (1 - 0.369) = 0.456 / 0.631 = 0.72 \rightarrow \text{Substantial agreement}$$

Conclusion: Helpfulness annotations are reliable — $\kappa > 0.6$ threshold met.

Fleiss' Kappa — Multiple Annotators

$$\kappa_F = (\bar{P} - \bar{P}_e) / (1 - \bar{P}_e)$$

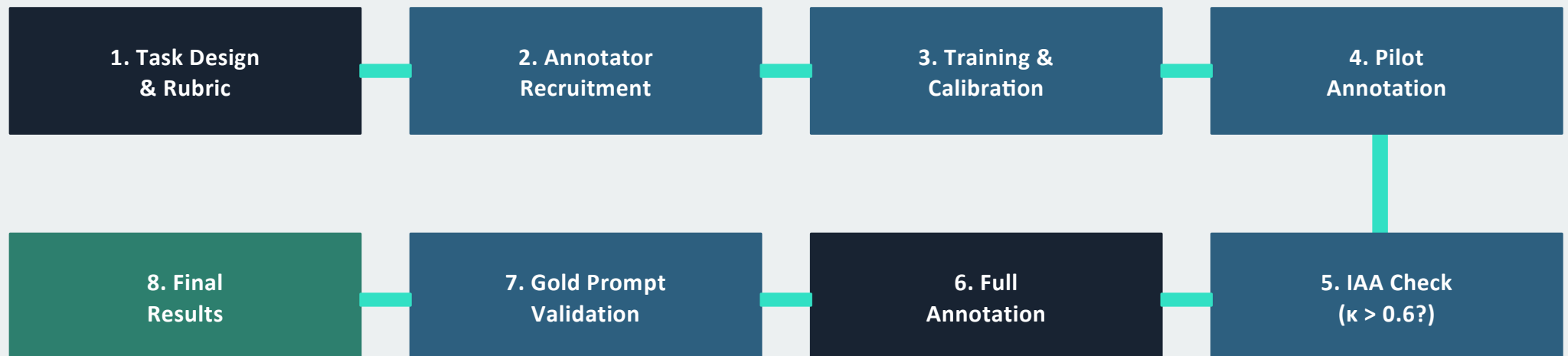
1. Generalises Cohen's κ to $k > 2$ annotators; each of N items rated by exactly n annotators
2. $\bar{P}$ = mean pairwise agreement across all (i,j) annotator pairs and all items
3. $\bar{P}_e$ = expected agreement by chance = $\sum_j p_j^2$ where p_j = proportion of all ratings in category j
4. Example: 3 annotators, 100 items, 3 categories (Low/Med/High). Category proportions: Low=0.25, Med=0.50, High=0.25
5. $\bar{P}_e = 0.25^2 + 0.50^2 + 0.25^2 = 0.375$. If $\bar{P}_{\text{observed}} = 0.72 \rightarrow \kappa_F = (0.72 - 0.375) / (1 - 0.375) = 0.55$

Note: Fleiss' κ decreases as annotator count increases — target $\kappa > 0.6$ for 3–5 annotators; $\kappa > 0.5$ acceptable for larger panels

Cohen's Kappa Interpretation Bands — LLM Evaluation Guidelines

κ Range	Interpretation	Action for LLM Eval
< 0.20	Slight agreement	Stop: re-design annotation task and instructions from scratch
0.20–0.40	Fair agreement	Pause: run calibration session with annotators; clarify rubric
0.40–0.60	Moderate agreement	Caution: acceptable for exploratory work; add more annotators
0.60–0.80	Substantial agreement	Good: suitable for production evaluation; standard reporting
0.80–1.00	Almost perfect	Excellent: task likely too easy; consider harder annotation challenges

Human Evaluation Workflow — Quality-Assurance Pipeline



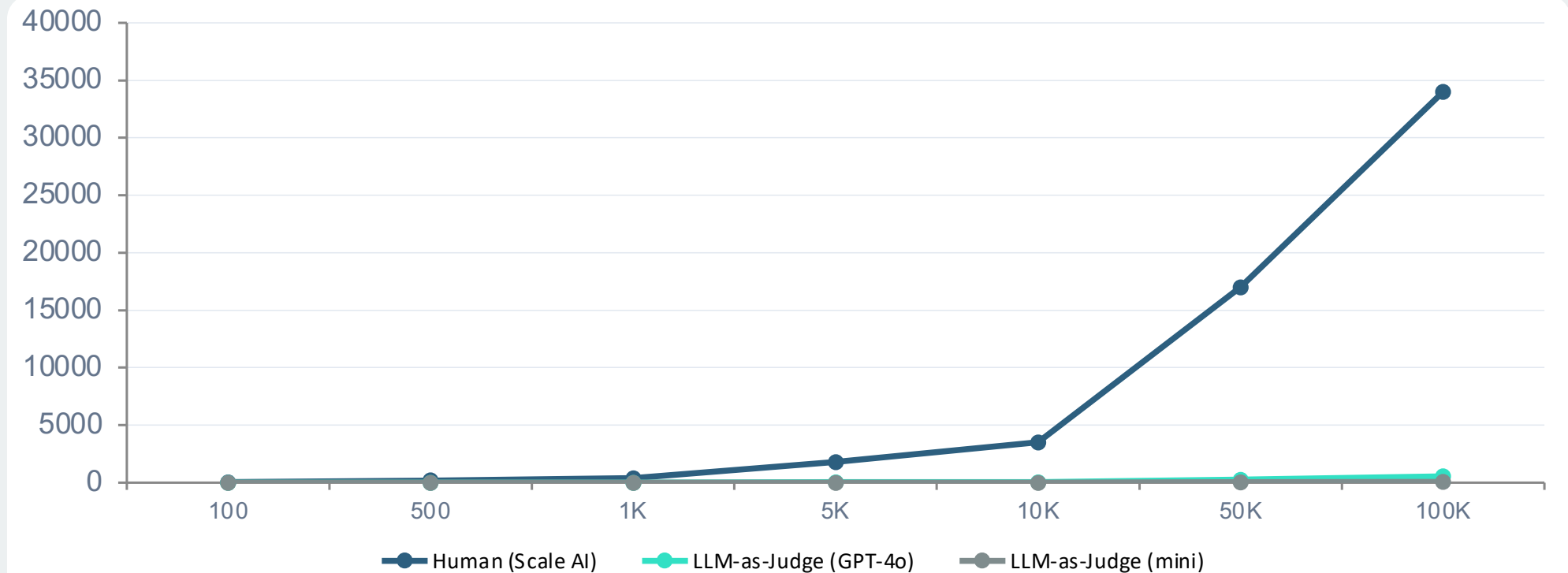
$\kappa < 0.6 \rightarrow$ Return to Step 3

Budget guideline: ~10 hours total per 200-item annotation set (incl. design, training, QA, and reporting).

Annotation Platforms — Overview

Platform	Annotator Type	Cost / Task	Quality Controls	Best For
Scale AI	Trained specialists	\$0.05–0.50	Gold standards, expert review	High-stakes, safety eval
Surge AI	US-based workers	\$0.03–0.20	Qualification tests, audits	General NLP annotation
Prolific	Academic crowd panel	\$0.05–0.30	Demographic targeting	Research studies
MTurk	General crowd	\$0.01–0.10	HIT qualification, attention	High-volume, low-risk
Labelbox	In-house / ML teams	SaaS fee	Custom workflows	Enterprise, internal teams
Argilla (OSS)	Internal teams	Free	Custom quality plugins	Self-hosted, privacy-first

Annotation Cost vs Volume — Break-Even Analysis



Human eval: ~\$0.34/evaluation (Scale AI trained workers). GPT-4o judge: ~\$0.0055/evaluation. For decisions that matter, human eval is 60× more expensive — but provides higher validity for novel and safety-critical tasks.

Chatbot Arena — Live Crowdsourced Evaluation

Chatbot Arena (LMSYS, 2023–present)

Chatbot Arena is a live evaluation platform where users submit real queries to anonymous model pairs, compare responses side-by-side, and vote for the better response. Votes are aggregated into an ELO-style rating system that produces a leaderboard correlating strongly with expert evaluation.

- Over 2 million human preference votes as of 2025 — the largest LLM preference dataset in existence
- Blind evaluation: users don't know which model produced which response — eliminates brand bias
- ELO rating system (developed for chess) provides calibrated, relative rankings with uncertainty intervals
- Covers diverse real-world queries: not researcher-designed tasks — reflects actual user intent distribution
- Pearson $r = 0.97$ with AlpacaEval 2.0-LC — validated as the reference human preference leaderboard

Chatbot Arena — Top Model ELO Ratings (May 2026)

ELO ratings are relative — a 100-point difference implies ~64% win rate for the higher-ranked model

1320

o3 (OpenAI)
#1 overall — May 2026

1295

Claude 3.7 Sonnet
#2 — Anthropic

1281

Gemini 2.0 Ultra
#3 — Google DeepMind

2M+

Total Human Votes
(cumulative 2023–2026)

Key Takeaways

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- 1 Human evaluation is irreplaceable. it is the only evaluation that directly measures whether outputs are useful and safe for real people
- 2 Pairwise preference studies (A/B voting) produce more reliable annotations than absolute Likert scales — use them for model selection decisions
- 3 Always compute inter-annotator agreement (Cohen's κ or Fleiss' κ) before trusting human labels; reject annotation batches with $\kappa < 0.40$
- 4 $\kappa > 0.60$ (substantial) is the minimum threshold for evaluation data used in production model decisions or RLHF training
- 5 Chatbot Arena's 2M+ live human votes make it the gold-standard model ranking platform — its ELO scores have $r=0.97$ with expert preference